543 Readmissions Project Exploratory Data Analysis Report

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Contents

1 Dataset Overview

The dataset comes from two CSV files, `healthcare_readmissions_dataset_train` and `healthcare_readmissions_dataset_test`, each row consisting of one patient encounter. The training file contains 8,038 rows and 19 columns (including the target variable).

Variable	Description
PatientID	Unique patient identifier
Age	Patient age
Gender	String variable with values "Male" or "Female"
Ethnicity	String variable with values "Caucasian", "Hispanic", "African American" and "Other"
Hospital ID	Hospital identifier with values "Hosp1", "Hosp2", and "Hosp3"
Height	Patient height in meters
Smoker	Boolean indicating if patient is a current smoker
BMI	Patient Body Mass Index
Weight	Patient weight in kg
Adjusted Weight	Health system-specific adjustments to patient weight (in kg)
Has Diabetes	Boolean indicating if patient has diabetes (0 or 1)
Has Hypertension	Boolean indicating if patient has hypertension (0 or 1)
Has Chronic Kidney Disease	Boolean indicating if patient has chronic kidney disease (0 or 1)
Living Situation	Living arrangement: Lives Alone, Lives with Family, Assisted Living
Exercise Frequency	String variable with values "None", "Occasional", or "Regular"
Diet Type	String variable with values "Balanced", "High-fat", "Vegetarian", "Other"
Number of Prior Visits	Number of previous hospitalizations of the patient
Medications Prescribed	Number of different prescription medications patient is currently taking
Length of Stay	Length of the hospital stay in days
Type of Treatment	String variable with values "None", "Minor Surgery", "Major Surgery", "Other Treatment"
Readmission within 30 days	Binary variable (1 = patient was readmitted, 0 = patient was not readmitted)
Post-Discharge Satisfaction Score	Survey done 30 days after discharge. 1.0 – 5.0 scale

Figure 1: Data Dictionary

It is worth noting that `Has Chronic Kidney Disease`, `Post-Discharge Satisfaction Score`, and `Living Situation` are not actually present in the files, despite being in the dictionary.

2 Data Quality Summary

Only two features in the dataset have missing values: `Number of Prior Visits` (314 rows or 3.9%) and `Medications Prescribed` (657 rows or 8.2%). The remaining features are complete, with no duplicates in the training set either. Although boolean columns are not consistent in datatype (`Smoker` is a boolean column while `Has Diabetes` and `Has Hypertension` are integers with values of 0 and 1), this can be handled quite easily during preprocessing.

`Exercise Frequency` and `Type of Treatment` also feature 'None' as a possible value, which the `pandas` function `pd.read_csv()` will interpret as a `NaN` by default. Again, this is not a major issue, just something to be aware of.

A glance at the distribution of variables shows that `Age` has problematic outliers, in the form of patients over the age of 100.

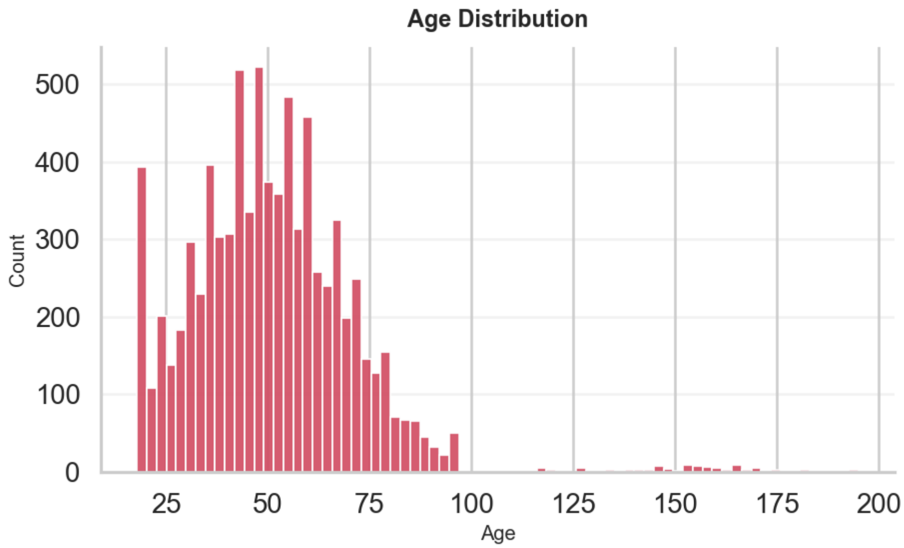

Figure 2: Raw Age Distribution

Using z-scores to identify outliers (as well as common sense), 80 rows were determined to be outliers and subsequently removed from the dataset. These rows make up 0.99% of the training set; although I would prefer to keep as much data as possible, these rows are clearly unrepresentative of real patients (for whatever reason), and could potentially skew the results of any analysis or modeling.

3 Statistical Summary

All stats are derived from the training set with the age outliers removed as outlined in the previous section.

3.1 Numeric Statistics

Table 1: Measure Summary Statistics

	count	mean	std	min	25%	50%	75%	max
age	7958.000000	50.110000	17.300000	18.000000	38.000000	50.000000	62.000000	95.000000
height_m	7958.000000	1.700000	0.100000	1.300000	1.600000	1.700000	1.800000	2.000000
bmi	7958.000000	26.260000	4.750000	8.300000	23.100000	26.200000	29.300000	44.000000
weight_kg	7958.000000	77.150000	18.950000	23.300000	64.800000	75.400000	87.300000	236.300000
adjusted_weight_kg	7958.000000	76.270000	16.660000	23.130000	64.730000	75.130000	86.860000	159.050000
has_diabetes	7958.000000	0.130000	0.340000	0.000000	0.000000	0.000000	0.000000	1.000000
has_hypertension	7958.000000	0.170000	0.380000	0.000000	0.000000	0.000000	0.000000	1.000000
number_of_prior_visits	7646.000000	3.040000	1.740000	0.000000	2.000000	3.000000	4.000000	11.000000
medications_prescribed	7308.000000	3.510000	1.960000	0.000000	2.000000	3.000000	5.000000	12.000000
length_of_stay	7958.000000	2.540000	2.970000	0.000000	0.000000	2.000000	4.000000	23.000000
readmission_within_30_days	7958.000000	0.170000	0.380000	0.000000	0.000000	0.000000	0.000000	1.000000

3.2 Categorical Statistics

Figure 3: Category Distributions

Individual distributions can be found in the **Univariate Analysis** section below.

4 Univariate Analysis

4.1 Numeric Features

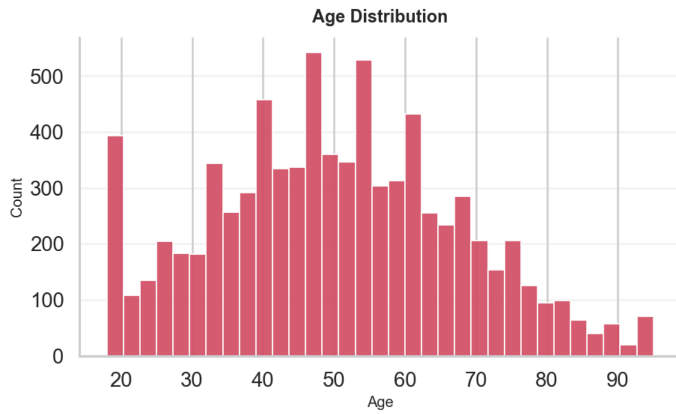

Figure 4: Age Histogram

The distribution with removed outliers makes much more sense compared to the raw values.

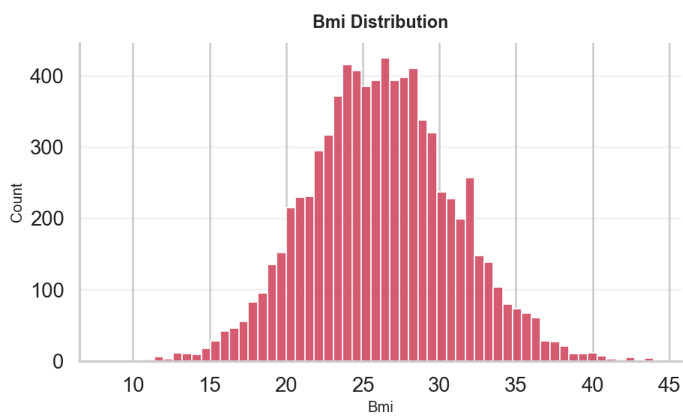

Figure 5: BMI Histogram

For the most part normal; some values on the ends are in the extreme but not necessarily impossible (e.g. a BMI of 44 is very high but not unheard of).

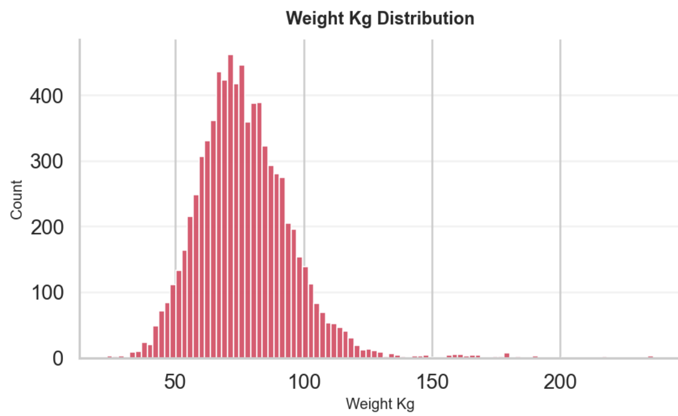

Figure 6: Weight Histogram

Similar to BMI, some extreme values but nothing that stands out as impossible. Also makes sense that its similar to BMI as they are highly correlated.

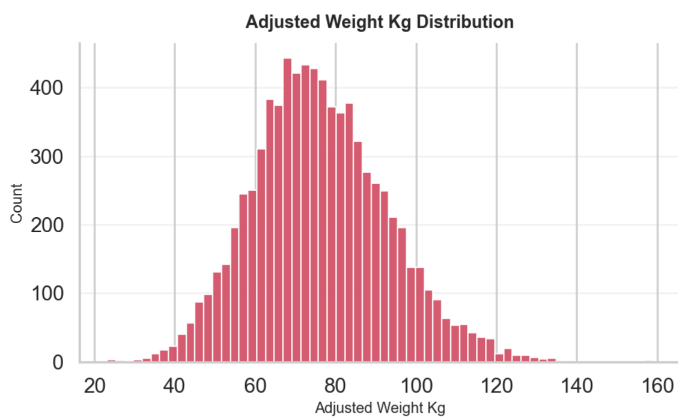

Figure 7: Adjusted Weight Histogram

Less outliers than standard weight, however the Data Dictionary does not specify how (or why) its adjusted, so it's hard to make any judgments or comparisons to standard weight.

4.2 Categorical Features

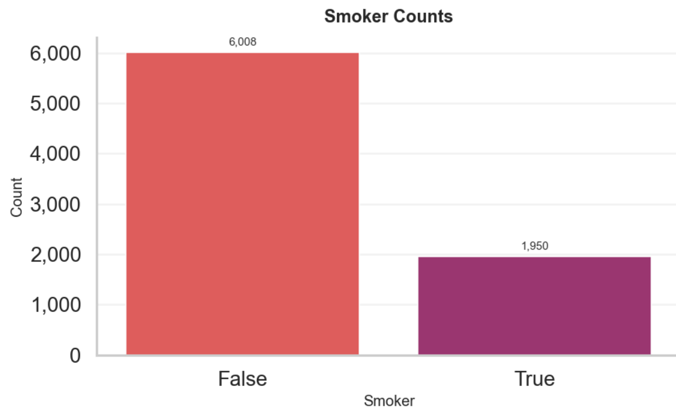

Figure 8: Smoker Bar Chart

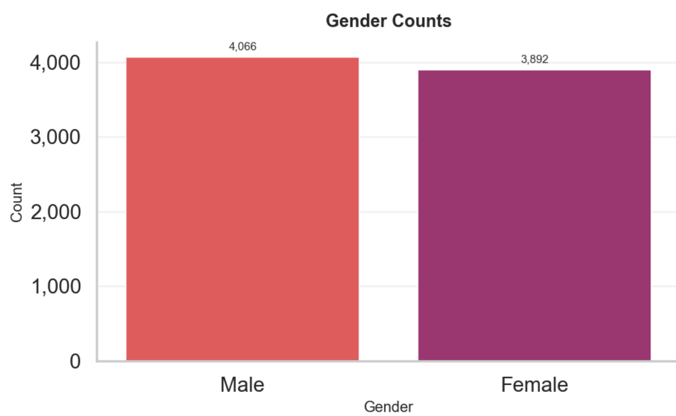

Figure 9: Gender Bar Chart

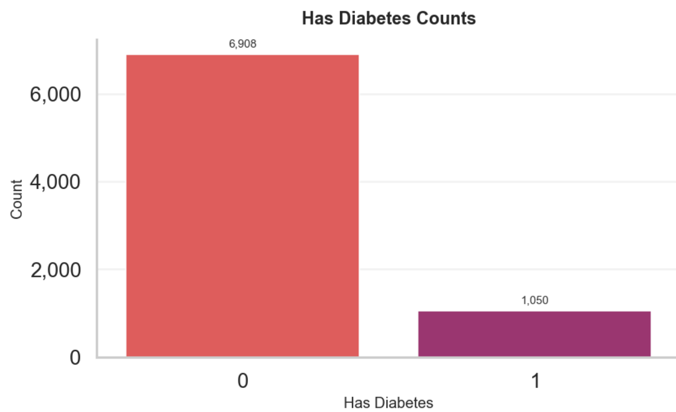

Figure 10: Diabetes Bar Chart

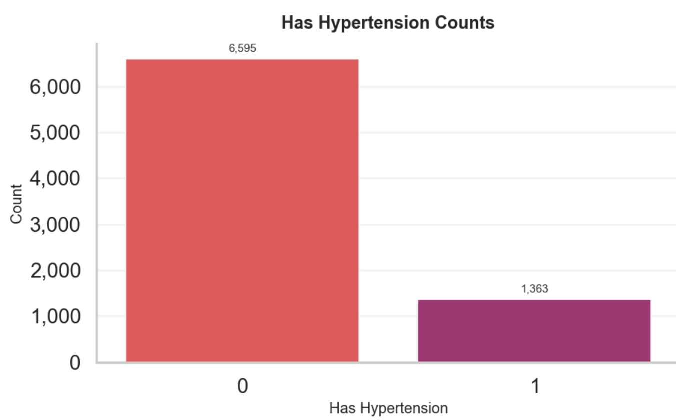

Figure 11: Hypertension Bar Chart

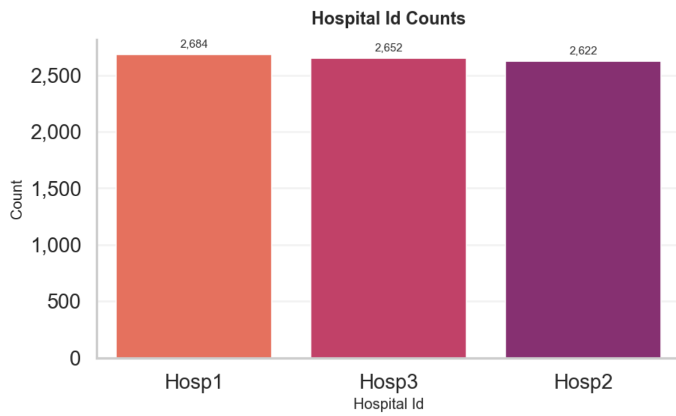

Figure 12: Hospital ID Bar Chart

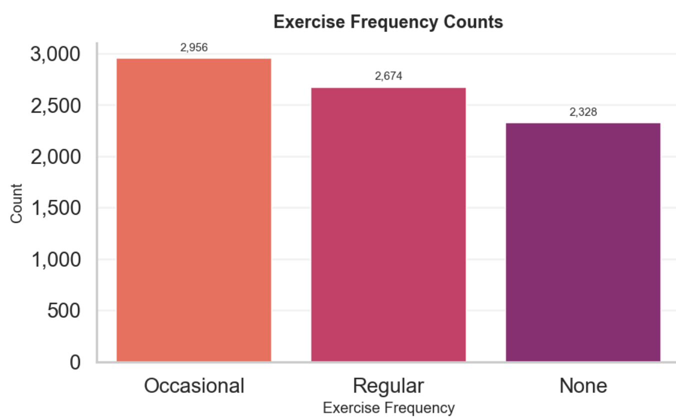

Figure 13: Exercise Frequency Bar Chart

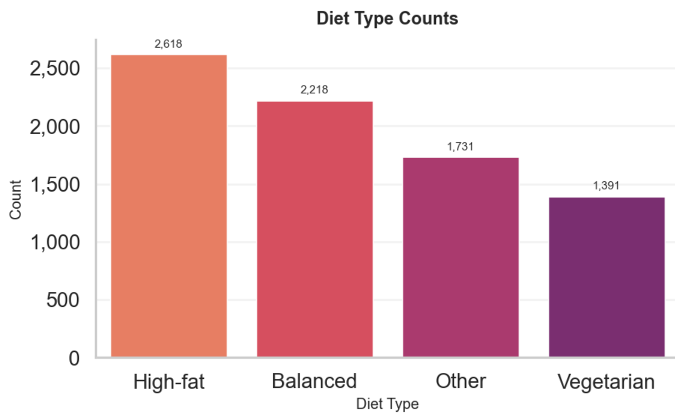

Figure 14: Diet Bar Chart

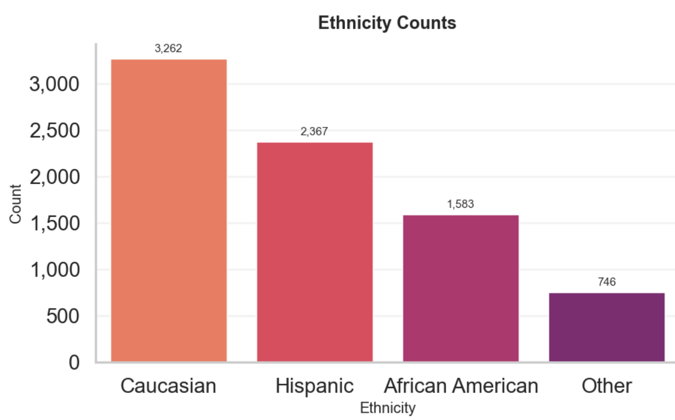

Figure 15: Ethnicity Bar Chart

Figures 8 through 15 are all quite unremarkable and to be expected for the dataset.

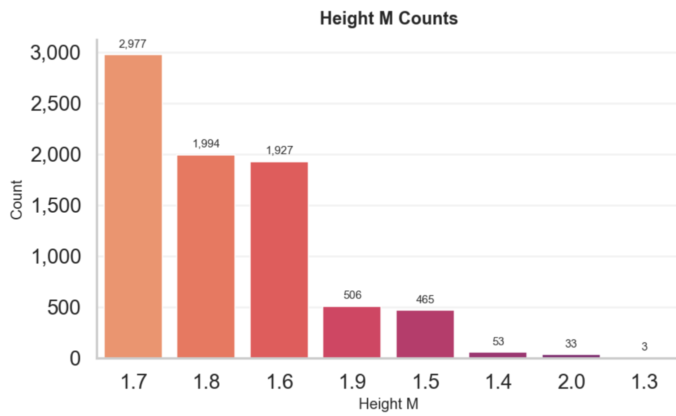

Figure 16: Height Bar Chart

Also expected, although interesting that the height is stored in what is essentially categorical form, rather than more precise measurements.

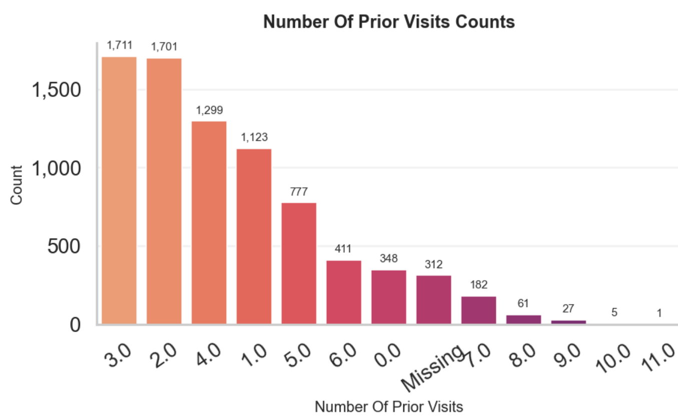

Figure 17: Prior Visits Bar Chart

Has a very long tail, and also missing values are quite prevalent. Would potentially make sense to group/encode values somehow, but that will be looked into during modeling and feature engineering.

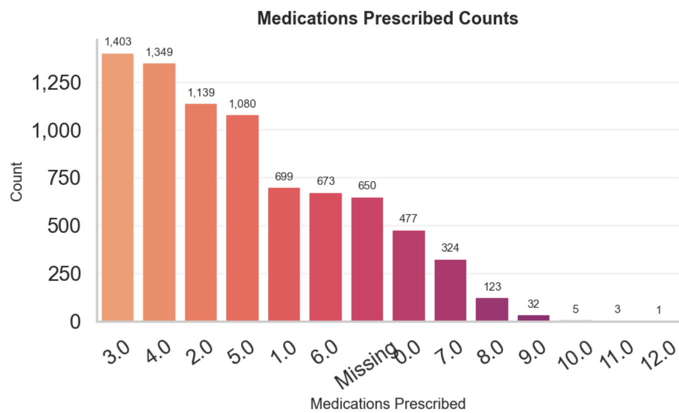

Figure 18: Medications Prescribed Bar Chart

A somewhat similar situation to prior visits, although an overall smaller range of values. Maybe worth grouping into ranges, or just full binary of 0 vs. 1 or more prescriptions.

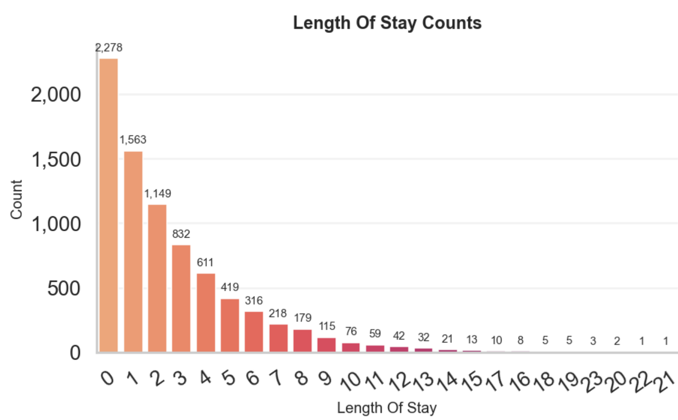

Figure 19: Length of Stay Bar Chart

Long tail, no missing values. Should be encoded somehow, should look to common techniques other projects/models have used in the past as it seems like a common and important feature in healthcare datasets.

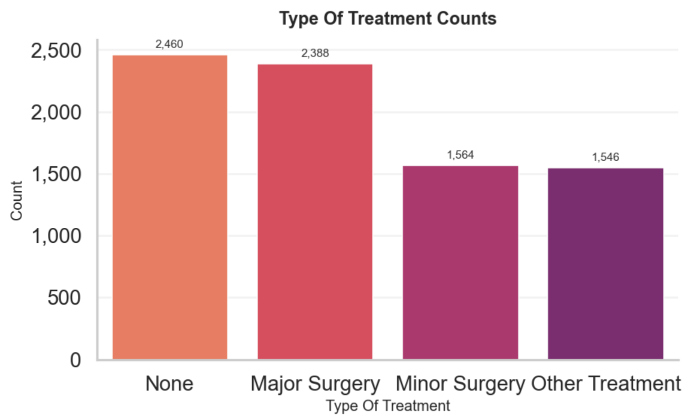

Figure 20: Treatment Type Bar Chart

Somewhat vague categories, more experimentation on encoding strategies should be revealing as per best utilizing the feature.

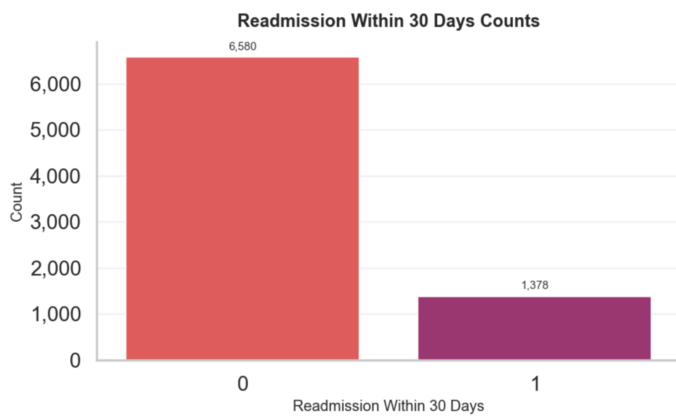

Figure 21: Readmission (Target) Bar Chart

Output variable. Very skewed, should consider that in model selection and/or resambling techniques.

5 Bivariate Analysis

5.1 Feature vs. Target

5.1.1 Categories

Figure 22: Categories vs. Readmission Bar Charts

Individual charts can be found in Individual Category vs. Readmission Charts. Across most, the ratio of readmissions to non-readmissions is fairly consistent, however the features like `Smoker` and `Has Diabetes` have a higher proportion of positive readmissions when the feature is present (i.e. smoker vs. non-smoker, has diabetes vs. does not have diabetes). Based off of limited domain knowledge this makes sense, mostly it is just something to be aware of when considering feature importance.

5.1.2 Measures

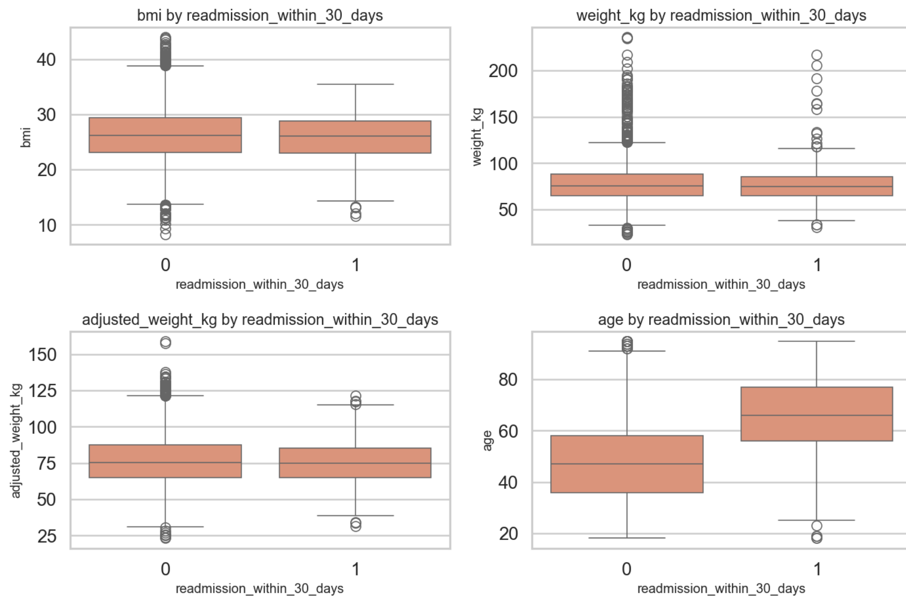

Figure 23: Measures vs. Readmission Box Plots

The most noticable distribution difference comes in age, with those who are readmitted being older on average than those who are not. This seems pretty intuitive, and there is not much to say about the others

5.2 Feature Correlations

Categorical features were one-hot encoded in order to include the entire dataset in the matrix.

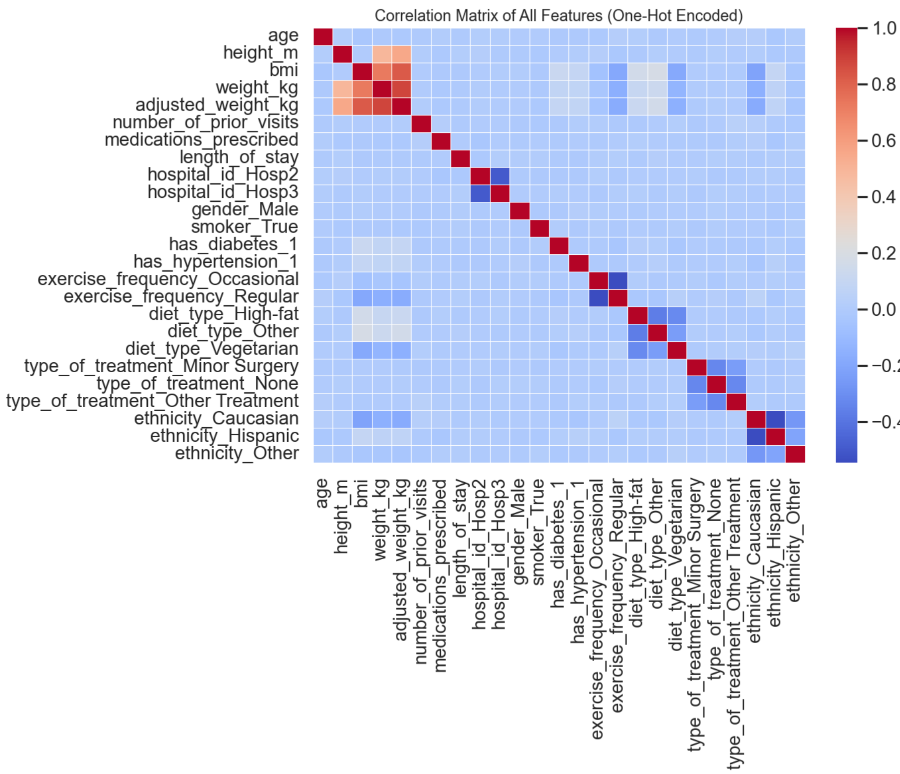

Figure 24: Correlation Matrix

Appendix A Individual Category vs. Readmission Charts

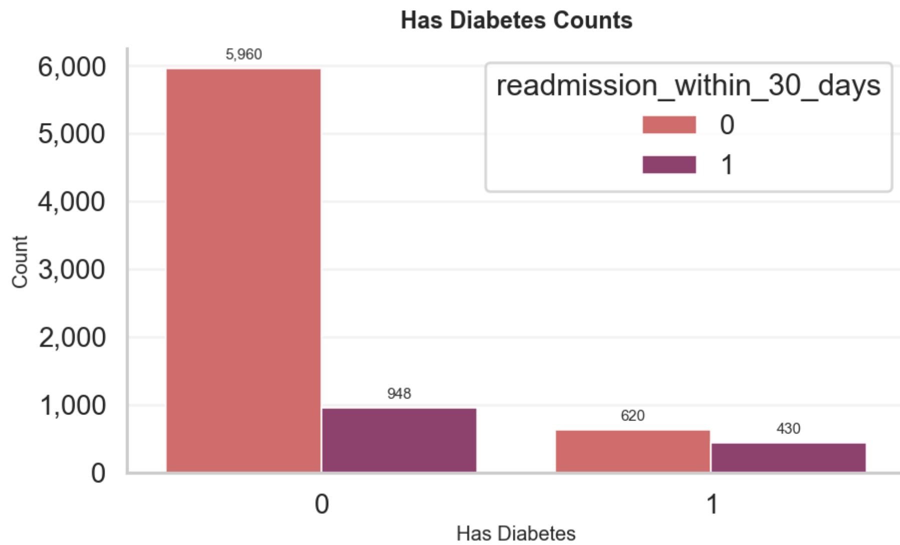

Figure 25: Diabetes vs. Readmission

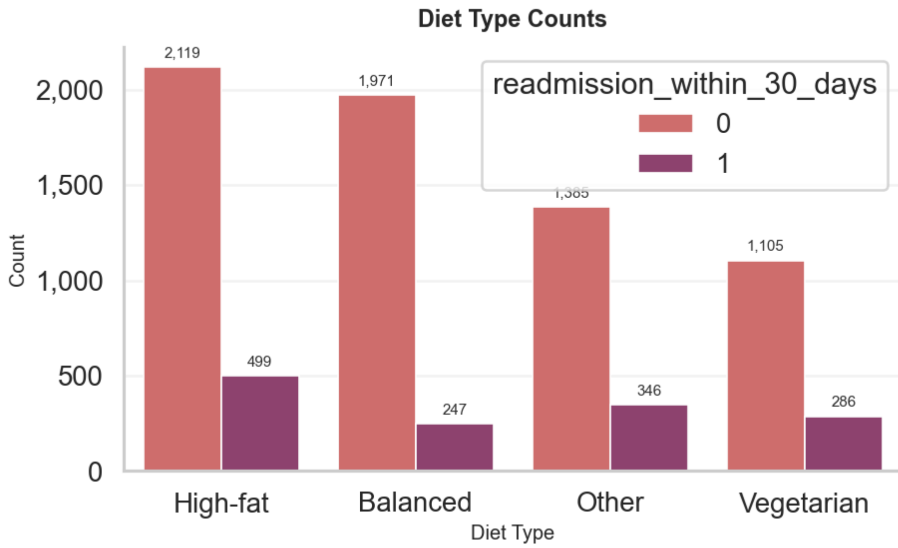

Figure 26: Diet vs. Readmission

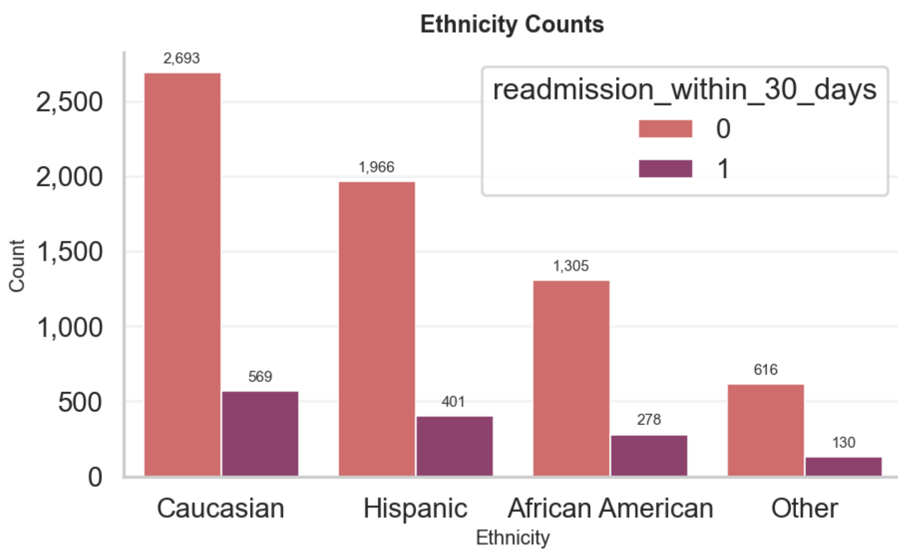

Figure 27: Ethnicity vs. Readmission

Figure 28: Exercise vs. Readmission

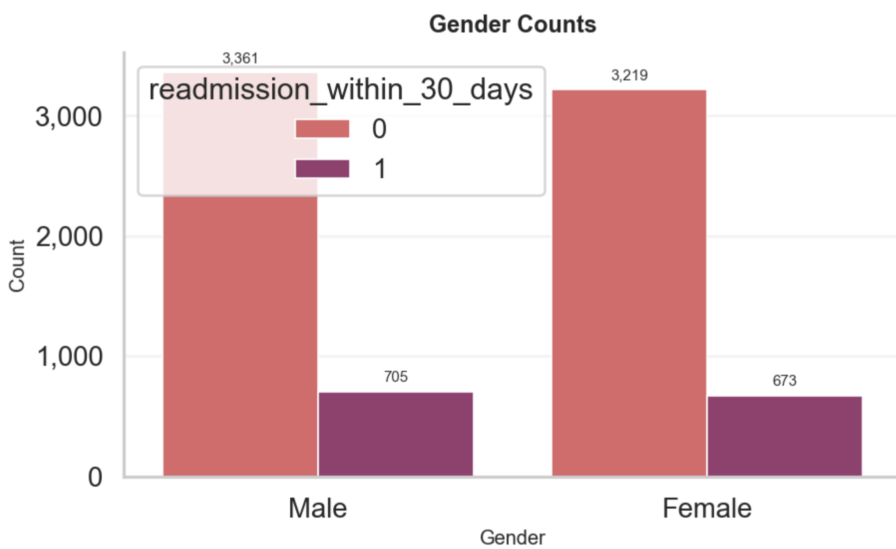

Figure 29: Gender vs. Readmission

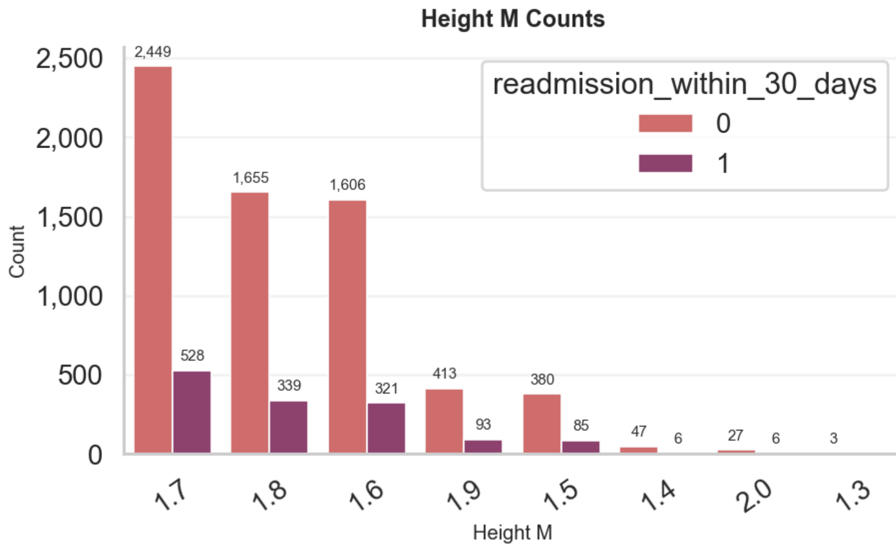

Figure 30: Height vs. Readmission

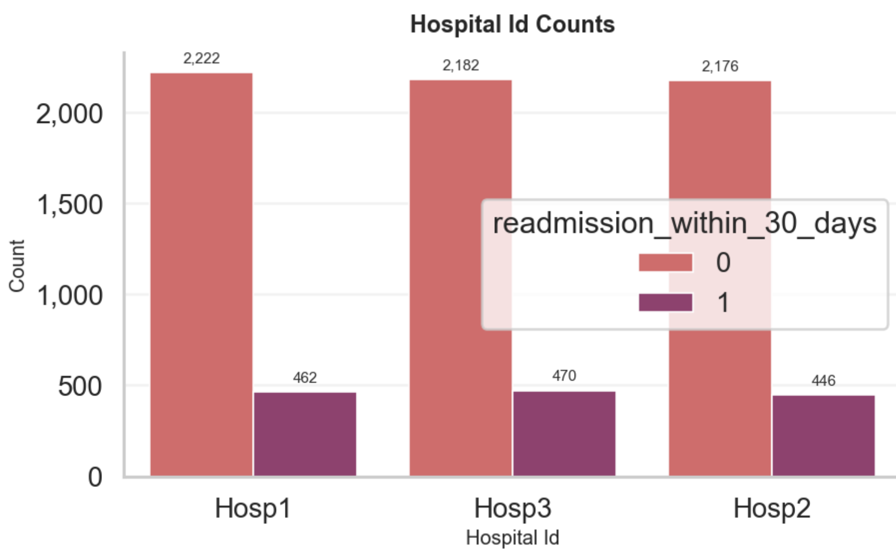

Figure 31: Hospital ID vs. Readmission

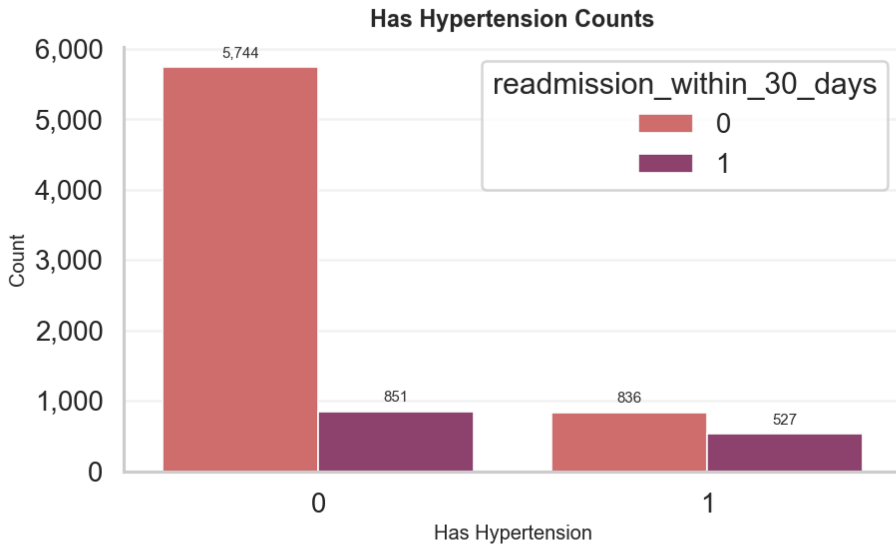

Figure 32: Hypertension vs. Readmission

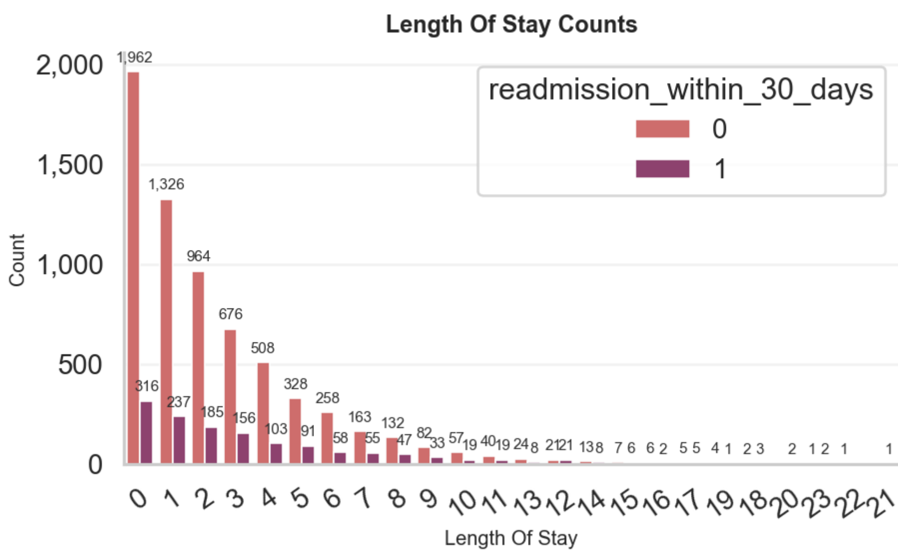

Figure 33: Length of Stay vs. Readmission

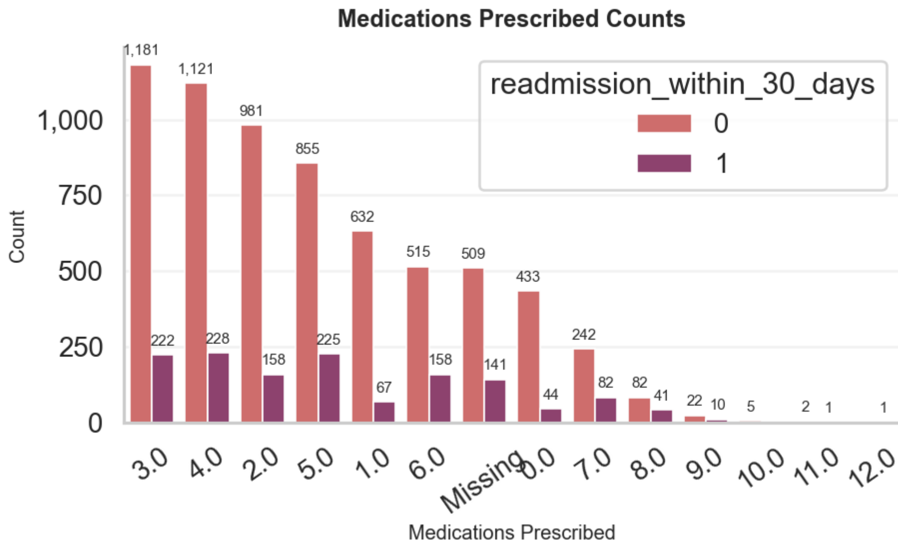

Figure 34: Num Medications vs. Readmission

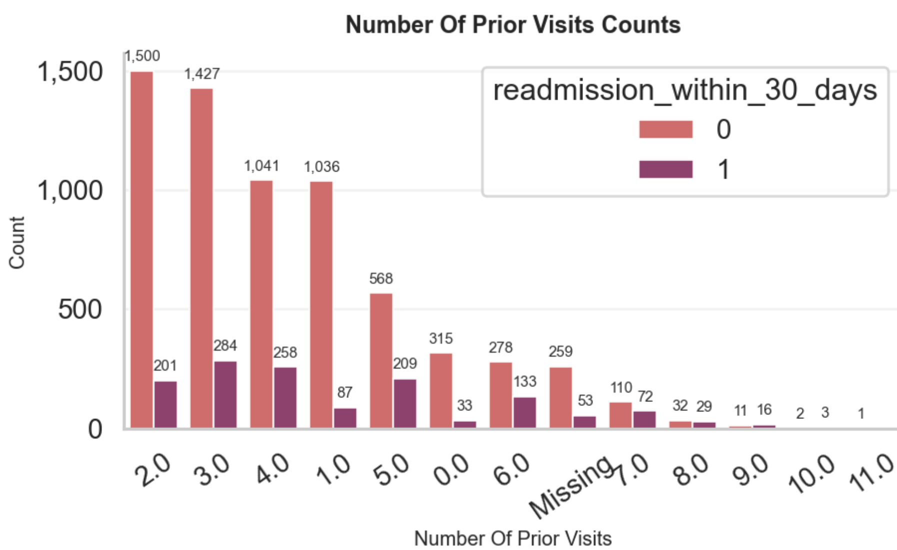

Figure 35: Num Prior Visits vs. Readmission

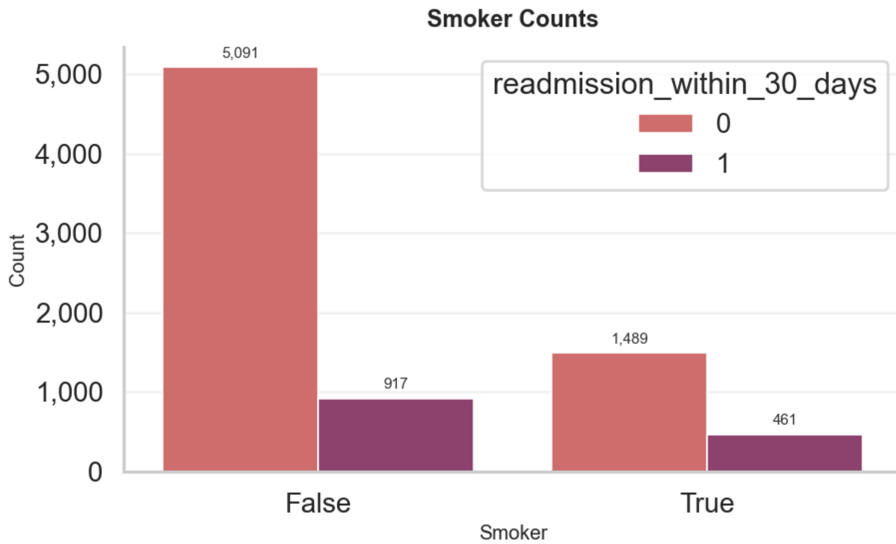

Figure 36: Smoker vs. Readmission

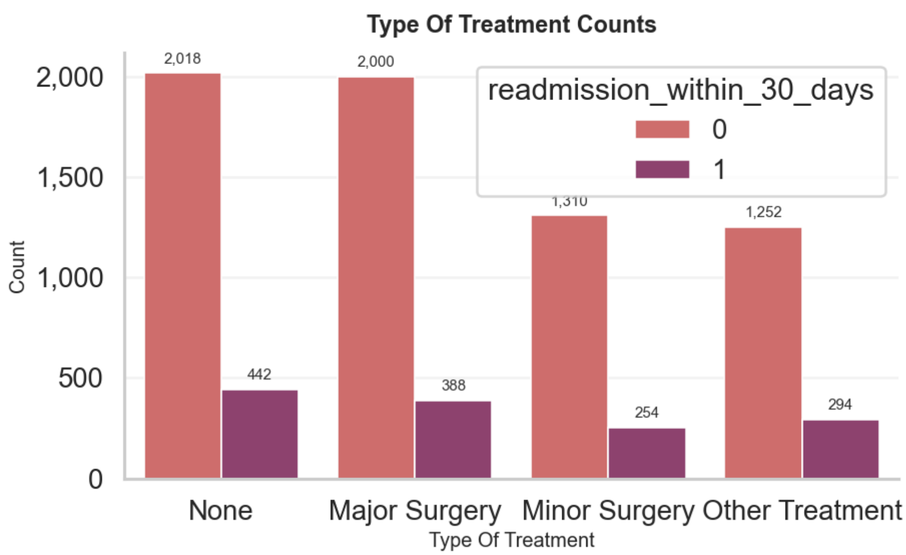

Figure 37: Type of Treatment vs. Readmission